

Intermediate DSA Coding Practice — Questions #526–#550

25 intermediate-level coding problems with metadata, explanations, visible tests, and hidden tests.

#526. Daily Temperatures

Difficulty	Intermediate
Category	Stacks / Arrays
Slug	daily-temperatures
Tags	monotonic-stack, arrays

Problem Statement

Return, for each day, how many days must pass before a warmer temperature occurs; return 0 if none exists.

Constraints

$1 \leq N \leq 2 \times 10^5$; temperatures are integers from -50 to 100.

Input Format

N; then N temperatures.

Output Format

N waiting times.

Approach / Explanation

Use a decreasing stack of indices. Resolve previous days whenever the current temperature is warmer.

Visible Test Cases

#	Input	Expected Output
1	8 73 74 75 71 69 72 76 73	1 1 4 2 1 1 0 0
2	4 30 30 30 30	0 0 0 0
3	5 90 80 70 60 100	4 3 2 1 0

Hidden Test Cases

#	Input	Expected Output
1	6 60 62 61 65 64 70	1 3 1 2 1 0
2	3 100 99 98	0 0 0

#527. Asteroid Collision

Difficulty	Intermediate
Category	Stacks / Simulation
Slug	asteroid-collision
Tags	stack, simulation

Problem Statement

Positive asteroids move right and negative asteroids move left. Resolve collisions according to size and return the survivors.

Constraints

$1 \leq N \leq 10^5$; $1 \leq \text{abs}(A[i]) \leq 10^4$.

Input Format

N; then N asteroid values.

Output Format

Surviving asteroid values, or EMPTY.

Approach / Explanation

Use a stack. A collision can occur only when the stack top is positive and the incoming asteroid is negative.

Visible Test Cases

#	Input	Expected Output
1	5 5 10 -5 8 -8	5 10
2	3 8 -8 2	2
3	3 10 2 -5	10

Hidden Test Cases

#	Input	Expected Output
1	4 -2 -1 1 2	-2 -1 1 2
2	6 10 2 -5 -20 15 -15	-20

#528. Product Except Self

Difficulty	Intermediate
Category	Arrays / Prefix Products
Slug	product-except-self
Tags	prefix-product, arrays

Problem Statement

For every position, return the product of all array elements except the element at that position, without division.

Constraints

$2 \leq N \leq 10^5$; $-30 \leq A[i] \leq 30$.

Input Format

N; then N integers.

Output Format

N resulting integers.

Approach / Explanation

Build prefix products into the answer and multiply suffix products during a reverse pass.

Visible Test Cases

#	Input	Expected Output
1	4 1 2 3 4	24 12 8 6
2	3 -1 1 0	0 0 -1
3	4 2 3 4 5	60 40 30 24

Hidden Test Cases

#	Input	Expected Output
1	5 0 1 2 0 4	0 0 0 0 0
2	3 -2 -3 -4	12 8 6

#529. Subarray Sum Equals K

Difficulty	Intermediate
Category	Arrays / Prefix Sum
Slug	subarray-sum-equals-k
Tags	prefix-sum, hashmap

Problem Statement

Count contiguous subarrays whose sum equals K. Values may be negative or zero.

Constraints

$1 \leq N \leq 2 \times 10^5$; $-10^4 \leq A[i] \leq 10^4$.

Input Format

N K; then N integers.

Output Format

Number of qualifying subarrays.

Approach / Explanation

Store frequencies of previous prefix sums. For current prefix P, add the frequency of P-K.

Visible Test Cases

#	Input	Expected Output
1	3 3 1 2 1	2
2	3 0 0 0 0	6
3	5 5 1 2 3 2 5	2

Hidden Test Cases

#	Input	Expected Output
1	6 -2 -1 -1 2 -2 3 -3	4
2	4 10 1 2 3 4	0

#530. Three Sum Closest

Difficulty	Intermediate
Category	Arrays / Two Pointers
Slug	three-sum-closest
Tags	sorting, two-pointers

Problem Statement

Find the sum of three distinct elements closest to target. If tied, return the smaller sum.

Constraints

$3 \leq N \leq 2000$; $-10^4 \leq A[i] \leq 10^4$.

Input Format

N target; then N integers.

Output Format

Closest sum.

Approach / Explanation

Sort the array, fix one value, and use two pointers to search the remaining range.

Visible Test Cases

#	Input	Expected Output
1	4 1 -1 2 1 -4	2
2	3 5 1 2 3	6
3	5 0 -2 -1 1 4 5	0

Hidden Test Cases

#	Input	Expected Output
1	6 10 1 2 4 8 16 32	14
2	5 -7 -10 -3 0 2 8	-8

#531. Search Rotated Sorted Array

Difficulty	Intermediate
Category	Binary Search
Slug	search-rotated-sorted-array
Tags	binary-search, rotated-array

Problem Statement

Search for a target in a strictly increasing array rotated at an unknown pivot. Return its index or -1.

Constraints

$1 \leq N \leq 2 \times 10^5$; all values are distinct.

Input Format

N target; then rotated array.

Output Format

Zero-based index or -1.

Approach / Explanation

At each midpoint, identify the sorted half and determine whether the target lies inside it.

Visible Test Cases

#	Input	Expected Output
1	7 0 4 5 6 7 0 1 2	4
2	5 3 5 1 2 3 4	3
3	4 9 6 7 8 1	-1

Hidden Test Cases

#	Input	Expected Output
1	8 5 6 7 8 9 1 2 3 5	7
2	1 4 4	0

#532. Find Peak Element

Difficulty	Intermediate
Category	Binary Search
Slug	find-peak-element
Tags	binary-search, arrays

Problem Statement

Find any index whose value is greater than its existing neighbors. Treat outside boundaries as negative infinity.

Constraints

$1 \leq N \leq 10^5$; adjacent values are distinct.

Input Format

N; then N integers.

Output Format

Any valid peak index.

Approach / Explanation

If $A[mid] < A[mid+1]$, a peak exists to the right; otherwise a peak exists at or left of mid.

Visible Test Cases

#	Input	Expected Output
1	4 1 2 3 1	2
2	5 1 2 1 3 5	4
3	1 7	0

Hidden Test Cases

#	Input	Expected Output
1	6 6 5 4 3 2 1	0
2	5 1 3 2 4 1	3

#533. Allocate Books

Difficulty	Intermediate
Category	Binary Search / Greedy
Slug	allocate-books
Tags	binary-search, greedy, partition

Problem Statement

Partition books in fixed order among M students to minimize the maximum pages assigned to one student. Every student must receive a non-empty group.

Constraints

$1 \leq N \leq 10^5$; $1 \leq M \leq N$; $\text{pages} \leq 10^9$.

Input Format

N M; then N page counts.

Output Format

Minimum possible maximum pages.

Approach / Explanation

Binary-search a page limit and greedily count how many students are needed under that limit.

Visible Test Cases

#	Input	Expected Output
1	4 2 12 34 67 90	113
2	5 3 10 20 30 40 50	60
3	3 4 1 2 3	-1

Hidden Test Cases

#	Input	Expected Output
1	5 2 10 20 30 40 50	90
2	6 3 5 17 100 11 20 30	117

#534. Aggressive Cows

Difficulty	Intermediate
Category	Binary Search / Greedy
Slug	aggressive-cows
Tags	binary-search, greedy

Problem Statement

Place C cows in given stalls so that the minimum distance between any two cows is as large as possible.

Constraints

$2 \leq N \leq 10^5$; positions are distinct and $\leq 10^9$.

Input Format

N C; then stall positions.

Output Format

Largest achievable minimum distance.

Approach / Explanation

Sort stalls and binary-search a candidate distance, greedily placing cows as early as possible.

Visible Test Cases

#	Input	Expected Output
1	5 3 1 2 4 8 9	3
2	6 4 0 3 4 7 10 12	3
3	3 2 5 10 15	10

Hidden Test Cases

#	Input	Expected Output
1	5 2 1 100 200 300 1000	999
2	4 4 1 2 3 10	1

#535. Number of Islands

Difficulty	Intermediate
Category	Graphs / Grid DFS
Slug	number-of-islands
Tags	grid, dfs, bfs

Problem Statement

Count connected components of land cells in a binary grid. Connections are vertical and horizontal only.

Constraints

Rows×columns≤10⁶; grid values are 0 or 1.

Input Format

R C; then R binary rows.

Output Format

Island count.

Approach / Explanation

Run DFS or BFS from every unvisited land cell and mark its entire component.

Visible Test Cases

#	Input	Expected Output
1	4 5 11000 11000 00100 00011	3
2	2 2 11 11	1
3	3 3 000 000 000	0

Hidden Test Cases

#	Input	Expected Output
1	5 6 101010 010101 101010 010101 101010	15
2	1 4 1111	1

#536. Rotting Oranges

Difficulty	Intermediate
Category	Graphs / Multi-source BFS
Slug	rotting-oranges
Tags	bfs, grid

Problem Statement

Each minute, rotten oranges infect adjacent fresh oranges. Return the time until all rot, or -1 if impossible.

Constraints

$1 \leq R, C \leq 500$.

Input Format

R C; then grid values 0,1,2.

Output Format

Minutes required or -1.

Approach / Explanation

Initialize BFS with all rotten cells and process simultaneous levels.

Visible Test Cases

#	Input	Expected Output
1	<pre>3 3 2 1 1 1 1 0 0 1 1</pre>	4
2	<pre>3 3 2 1 1 0 1 1 1 0 1</pre>	-1
3	<pre>1 2 2 1</pre>	1

Hidden Test Cases

#	Input	Expected Output
1	<pre>2 2 2 1 1 1</pre>	2
2	<pre>2 2 0 0 0 0</pre>	0

#537. Bipartite Graph Check

Difficulty	Intermediate
Category	Graphs / Coloring
Slug	bipartite-graph-check
Tags	graph, bfs, coloring

Problem Statement

Determine whether an undirected graph can be colored with two colors so that adjacent vertices have different colors.

Constraints

$1 \leq N \leq 10^5$; $M \leq 2 \times 10^5$.

Input Format

N M; then M edges using 0-based vertices.

Output Format

YES or NO.

Approach / Explanation

Color each component using BFS. A same-color edge proves the graph is not bipartite.

Visible Test Cases

#	Input	Expected Output
1	4 4 0 1 1 2 2 3 3 0	YES
2	3 3 0 1 1 2 2 0	NO
3	5 2 0 1 3 4	YES

Hidden Test Cases

#	Input	Expected Output
1	6 6 0 1 1 2 2 3 3 4 4 5 5 0	YES
2	1 0	YES

#538. Kruskal MST

Difficulty	Intermediate
Category	Graphs / Minimum Spanning Tree
Slug	kruskal-mst
Tags	mst, kruskal, dsu

Problem Statement

Find the total weight of a minimum spanning tree of a connected undirected weighted graph.

Constraints

$1 \leq N \leq 2 \times 10^5$; $M \leq 3 \times 10^5$.

Input Format

N M; then edges u v w.

Output Format

MST total weight.

Approach / Explanation

Sort edges by weight and use Disjoint Set Union to add only edges joining different components.

Visible Test Cases

#	Input	Expected Output
1	4 5 1 2 1 2 3 2 3 4 1 4 1 4 1 3 3	4
2	3 3 1 2 5 2 3 6 1 3 2	7
3	2 1 1 2 9	9

Hidden Test Cases

#	Input	Expected Output
1	5 7 1 2 3 1 3 1 2 3 2 2 4 4 3 4 6 4 5 2 3 5 5	8
2	4 3 1 2 1 2 3 2 1 3 3	-1

#539. Evaluate Division

Difficulty	Intermediate
Category	Graphs / Weighted DFS
Slug	evaluate-division
Tags	graph, dfs, ratios

Problem Statement

Given equations $a/b = \text{value}$, answer division queries. Return -1 if variables are disconnected.

Constraints

Equations $\leq 10^4$; values positive.

Input Format

E; equations; Q; queries.

Output Format

One result per query.

Approach / Explanation

Build a weighted bidirectional graph and multiply weights along a DFS path.

Visible Test Cases

#	Input	Expected Output
1	2 a b 2 b c 3 3 a c b a a e	6 0.5 -1
2	1 x y 4 2 x y y x	4 0.25
3	1 a b 2 1 a a	1

Hidden Test Cases

#	Input	Expected Output
1	3 a b 1.5 b c 2 c d 4 2 a d d a	12 0.083333

#540. Longest Common Subsequence

Difficulty	Intermediate
Category	Dynamic Programming / Strings
Slug	longest-common-subsequence
Tags	dp, strings

Problem Statement

Find the length of the longest common subsequence of two strings and print one valid subsequence.

Constraints

String lengths ≤ 2000 ; lowercase English letters.

Input Format

Two strings on separate lines.

Output Format

Length and one valid LCS.

Approach / Explanation

Use a DP table for lengths and backtrack from the final cell to reconstruct a subsequence.

Visible Test Cases

#	Input	Expected Output
1	abcde ace	3 ace
2	abc abc	3 abc
3	abc def	0

Hidden Test Cases

#	Input	Expected Output
1	AGGTAB GTXAYB	4 GTAB
2	aaaa aa	2 aa

#541. 0/1 Knapsack Selection

Difficulty	Intermediate
Category	Dynamic Programming / Knapsack
Slug	zero-one-knapsack-selection
Tags	dp, knapsack

Problem Statement

Choose each item at most once to maximize value under capacity W . Output maximum value and selected item indices.

Constraints

$N \leq 200$; $W \leq 10^4$.

Input Format

N W ; then N lines weight value.

Output Format

Maximum value, count, and indices.

Approach / Explanation

Use 0/1 knapsack DP and backtrack through decisions to recover selected items.

Visible Test Cases

#	Input	Expected Output
1	3 50 10 60 20 100 30 120	220 2 2 3
2	2 5 6 10 7 20	0 0
3	4 7 1 1 3 4 4 5 5 7	9 2 2 3

Hidden Test Cases

#	Input	Expected Output
1	5 10 2 6 2 10 6 12 5 8 4 7	27 3 1 2 5
2	1 3 3 9	0 0

#542. Matrix Chain Multiplication

Difficulty	Intermediate
Category	Dynamic Programming / Intervals
Slug	matrix-chain-multiplication
Tags	dp, matrix-chain

Problem Statement

Given dimensions of matrices multiplied in order, find the minimum scalar multiplication cost.

Constraints

$2 \leq \text{number of matrices} \leq 200$; $\text{dimensions} \leq 10^4$.

Input Format

N; then N+1 dimensions.

Output Format

Minimum multiplication cost.

Approach / Explanation

Let $dp[i][j]$ be the minimum cost for matrices i through j and try every split point.

Visible Test Cases

#	Input	Expected Output
1	4 10 30 5 60	4500
2	3 10 20 30	6000
3	5 40 20 30 10 30	26000

Hidden Test Cases

#	Input	Expected Output
1	6 5 10 3 12 5 50	1655
2	3 1 1 1	1

#543. Decode Ways

Difficulty	Intermediate
Category	Dynamic Programming / Strings
Slug	decode-ways
Tags	dp, strings

Problem Statement

Count valid decodings where 1=A through 26=Z. A leading zero is invalid.

Constraints

$1 \leq \text{length} \leq 10^5$; digits only.

Input Format

One digit string.

Output Format

Count modulo 1,000,000,007.

Approach / Explanation

Use rolling DP, considering valid one-digit and two-digit codes at each position.

Visible Test Cases

#	Input	Expected Output
1	12	2
2	226	3
3	06	0

Hidden Test Cases

#	Input	Expected Output
1	11106	2
2	10	1
3	27	1

#544. Palindromic Substrings

Difficulty	Intermediate
Category	Strings / Dynamic Programming
Slug	palindromic-substrings
Tags	strings, palindrome, dp

Problem Statement

Count all palindromic substrings, counting equal substrings at different positions separately.

Constraints

$1 \leq \text{length} \leq 3000$; lowercase English letters.

Input Format

One string.

Output Format

Number of palindromic substrings.

Approach / Explanation

Expand around every odd and even center and count successful expansions.

Visible Test Cases

#	Input	Expected Output
1	aaa	6
2	abc	3
3	abba	6

Hidden Test Cases

#	Input	Expected Output
1	racecar	10
2	aaaaa	15

#545. LRU Cache Operations

Difficulty	Intermediate
Category	Design / HashMap / Linked List
Slug	lru-cache-operations
Tags	lru-cache, hashmap, linked-list

Problem Statement

Implement an LRU cache. GET returns a value or -1; GET and PUT make a key most recently used.

Constraints

Capacity $\leq 10^4$; operations $\leq 2 \times 10^5$.

Input Format

Capacity Q; then Q GET or PUT operations.

Output Format

Print GET results in order.

Approach / Explanation

Use a hashmap for lookup and a doubly linked list for recency ordering.

Visible Test Cases

#	Input	Expected Output
1	2 6 PUT 1 10 PUT 2 20 GET 1 PUT 3 30 GET 2 GET 3	10 -1 30
2	1 4 PUT 1 5 GET 1 PUT 2 6 GET 1	5 -1
3	2 3 GET 7 PUT 7 9 GET 7	-1 9

Hidden Test Cases

#	Input	Expected Output
1	2 7 PUT 1 1 PUT 2 2 GET 1 PUT 3 3 GET 2 GET 1 GET 3	1 -1 1 3

#546. Median of Two Sorted Arrays

Difficulty	Intermediate
Category	Binary Search / Arrays
Slug	median-two-sorted-arrays
Tags	binary-search, median

Problem Statement

Find the median of the combined sorted sequence of two sorted arrays in logarithmic time relative to the smaller array.

Constraints

Array lengths $\leq 10^5$; at least one array is non-empty.

Input Format

N M; then arrays A and B.

Output Format

Median as a decimal number.

Approach / Explanation

Binary-search a partition in the smaller array so left-side values do not exceed right-side values.

Visible Test Cases

#	Input	Expected Output
1	2 1 1 3 2	2.0
2	2 2 1 2 3 4	2.5
3	0 3 1 2 3	2.0

Hidden Test Cases

#	Input	Expected Output
1	3 4 -5 0 10 -2 4 6 8	4.0
2	1 0 7	7.0

#547. Minimum Meeting Rooms

Difficulty	Intermediate
Category	Intervals / Heap
Slug	minimum-meeting-rooms
Tags	intervals, heap, sorting

Problem Statement

Find the minimum rooms needed for meetings. A room is reusable when one meeting ends exactly as another begins.

Constraints

$1 \leq N \leq 2 \times 10^5$; $0 \leq \text{start} < \text{end} \leq 10^9$.

Input Format

N; then N start/end pairs.

Output Format

Minimum rooms.

Approach / Explanation

Sort meetings by start and maintain a min-heap of active ending times.

Visible Test Cases

#	Input	Expected Output
1	3 0 30 5 10 15 20	2
2	4 1 5 2 6 3 7 8 9	3
3	2 1 2 2 3	1

Hidden Test Cases

#	Input	Expected Output
1	5 0 10 1 2 3 4 5 6 7 8	2
2	4 1 10 2 3 4 5 6 7	2

#548. Top K Frequent Elements

Difficulty	Intermediate
Category	HashMap / Heap
Slug	top-k-frequent-elements
Tags	hashmap, heap, frequency

Problem Statement

Return the K most frequent distinct elements. Break frequency ties by smaller numeric value.

Constraints

$1 \leq N \leq 2 \times 10^5$; $K \leq \text{number of distinct values}$.

Input Format

N K; then N integers.

Output Format

Elements ordered by frequency descending, then value ascending.

Approach / Explanation

Count frequencies and sort or heap-order distinct values using the required comparator.

Visible Test Cases

#	Input	Expected Output
1	6 2 1 1 1 2 2 3	1 2
2	5 2 4 4 5 5 6	4 5
3	4 1 -1 -1 2 3	-1

Hidden Test Cases

#	Input	Expected Output
1	8 3 5 5 4 4 4 3 3 2	4 3 5
2	7 2 1 2 2 3 3 3 1	3 1

#549. Insert and Merge Interval

Difficulty	Intermediate
Category	Intervals / Arrays
Slug	insert-and-merge-interval
Tags	intervals, merging

Problem Statement

Insert a new interval into sorted non-overlapping intervals and merge all overlaps.

Constraints

$0 \leq N \leq 10^5$; endpoints $\leq 10^9$.

Input Format

N; intervals; final new interval.

Output Format

Resulting intervals.

Approach / Explanation

Append intervals before the new interval, merge overlaps, then append remaining intervals.

Visible Test Cases

#	Input	Expected Output
1	3 1 3 6 9 12 15 2 7	2 1 9 12 15
2	2 1 2 5 6 3 4	3 1 2 3 4 5 6
3	0 4 8	1 4 8

Hidden Test Cases

#	Input	Expected Output
1	4 1 5 10 12 15 18 20 25 0 30	1 0 30
2	3 2 4 6 8 10 12 8 11	3 2 4 6 12